

User Experience & Activation Strategy

Who uses the park, when, and why the programme targets late afternoon in spring and autumn.

07

OF 12 UPLOAD SLOTS

64.6%

COMFORTABLE AS DESIGNED

+20.1 pts

COMFORT HOURS GAINED

4,402

DAYLIGHT HOURS MODELLED

97.5%

COMFORT BAND ACCURACY

▶ EVERY FIGURE IN THIS FILE CAN BE CHECKED HERE
wasim437.github.io/AI_SAFA/

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01

WHAT IS IN THIS FILE

Phase8 User Experience and Activation Report	DRAWING
Childrens dune play	RENDER
Circulation	DRAWING
Fig08 microclimate regimes	FIGURE
Fig09 diurnal comfort	FIGURE

02

THE SCALE OF THE ANALYSIS BEHIND IT

39 years

of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

8,760 hours

of sun position ray-traced against 131 trees for every hour of the year

15,000

one-metre ground cells modelled for shade and comfort, site-wide

4 models

trained and tested for leakage — two of them decide what this document reports

41 checks

run on every rebuild, so a wrong number fails loudly instead of shipping quietly

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

Produced by **CODE** [src/models.py](#) [src/climate.py](#) **DATA** [data/processed/hourly_climate_comfort_8760.csv](#)

Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

07

OF 12

01

THE TEN PHASES

P1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

P3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

P5 Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

P7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

P9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

P2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

P4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

P6 Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

P8 User Experience & Activation

THIS DOCUMENT

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

P10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

02

THE CODE AND DATA BEHIND THIS DOCUMENT

Models

src

Climate

src

Hourly climate comfort

data/processed

03

REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 41 checks
```

UPLOAD SLOT 07 OF 12 · USER EXPERIENCE & ACTIVATION STRATEGY

User Experience & Activation Strategy

Who uses this park, when, and what the climate permits

Activation strategies usually assert a programme. This one derives it. The comfort model says exactly when the park is pleasant, and the programme is built to match those windows rather than to fill a calendar — which means committing, in writing, to not programming summer midday outdoors.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 04 August 2026 · every quantity regenerated by python `run_analysis.py`

1. The catchment

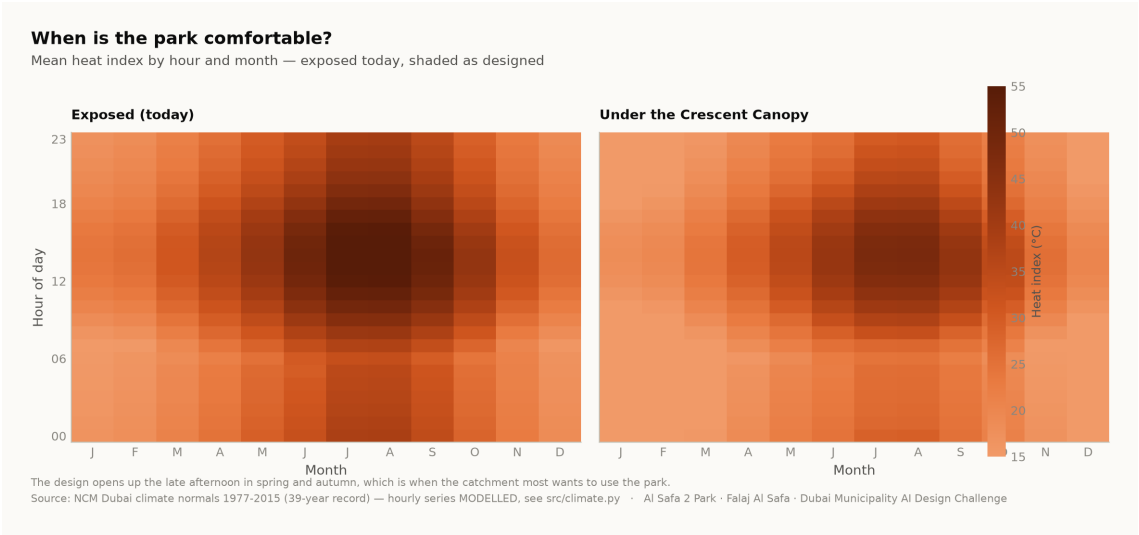
Approximately **7,640 residents** live within a ten-minute walk of the site (Dubai Statistics Center, 2023). Al Safa 2 is an established low-rise residential neighbourhood with a mixed demographic — families with young children, working adults, domestic staff, and a significant population of older residents.

2. Five user groups, and what each one needs

User group	When they come	What the design owes them
Families with young children	Late afternoon, weekends	Play in shade, seating with sightlines to it, a drinking fountain and a restroom within sight of the play area
Older residents	Early morning, evening	Step-free routes, shaded rest never far from the walk, seating at close intervals, no level change without a ramp
Runners and walkers	Dawn and after dark	A continuous loop with a known distance, lit at ankle height, separated from play and picnic
Workers on a break	Midday, year round	The one part of the site that is genuinely cool at noon — the crescent walk and the basin
Community and events	Evenings, cooler months	A flexible plaza and event lawn on the convex face, with the souk adjacent so trade and programme reinforce each other

3. When the park is actually comfortable

The comfort model resolves every hour of the year. Today 44.5% of daylight hours are comfortable; as designed, 64.6% are. The gain is not spread evenly, and that is the useful part: it concentrates in **late afternoon, in spring and autumn**.



Comfort by hour and by month. The design opens up the late afternoon in the shoulder seasons. Analysis output — computed from project data.

4. The activation calendar, derived from that

Season	Window	Programme
Oct – Apr (shoulder and winter)	16:00 – 21:00, and weekend mornings	Community events on the plaza and event lawn, evening souk trading, outdoor fitness classes, weekend family programming
May – Sep (summer)	Before 09:00 and after 18:00 only	Early running on Al Madar, evening use of the basin and the crescent walk, night programming on the plaza. Nothing is scheduled outdoors at midday.

Committing to a quiet summer midday is a design position, not a gap. Programming an outdoor event into a 56 °C heat index would be a scheduling decision that the analysis in this submission directly contradicts. The park is designed so that the hours it does claim are real.

5. Comfort, accessibility and inclusion

- The site is level, so the whole park is step-free. The only change of level in the scheme is the Oasis Basin, which is ramped.
- Every room is reached from the shaded primary route by a single alley — no room requires crossing another to arrive.
- Shaded rest (the majlis pods) is never far from the walk, so a user who needs to stop can always reach shade.
- 6 drinking fountains and 2 universally accessible restroom blocks are distributed against actual dwell points, not spaced evenly for the drawing.
- Play is divided by age group, with family seating in shade and clear sightlines to both zones.

6. Day and night

The brief asks for a park that works across the whole day. The convex face carries the uses that run after dark — the plaza, the event lawn and the souk — because its exposure stops being a liability once the sun is down. The crescent is lit from within so the arc reads at night as it does by day, and Al Madar is lit at ankle height for early and late running.

Childrens dune play

Artistic impression — illustrative of design intent. AI-generated.

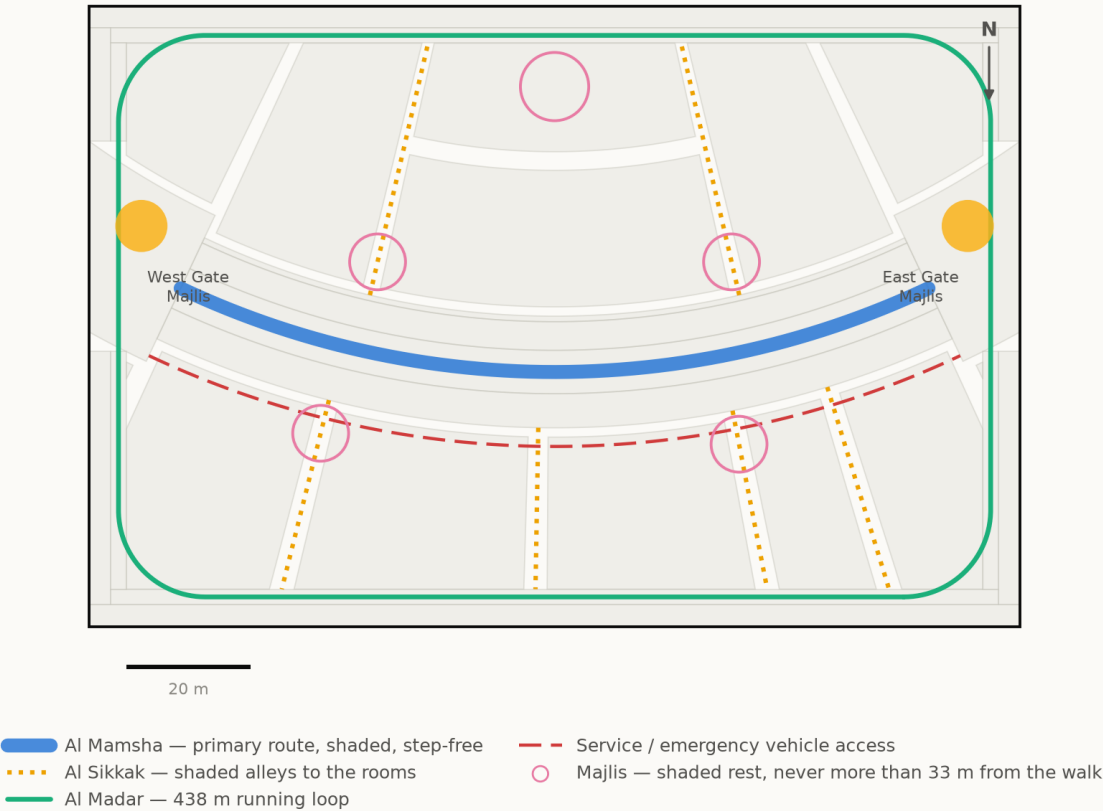


Circulation

Every route through the park, and the step-free claim it supports

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: [src/plan.py](#) — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

DRAWN BY [src/drawings.py](#) [src/plan.py](#)

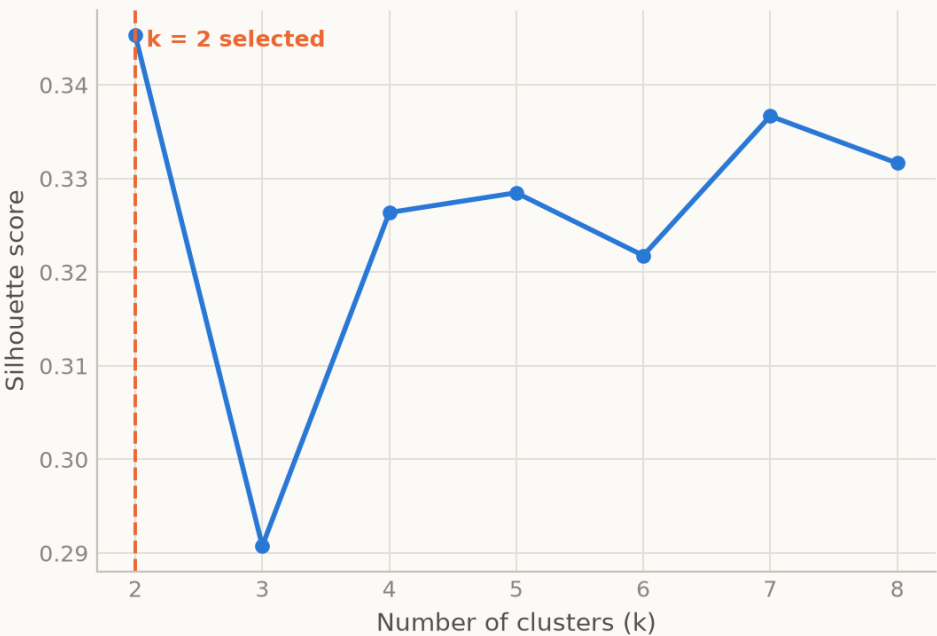
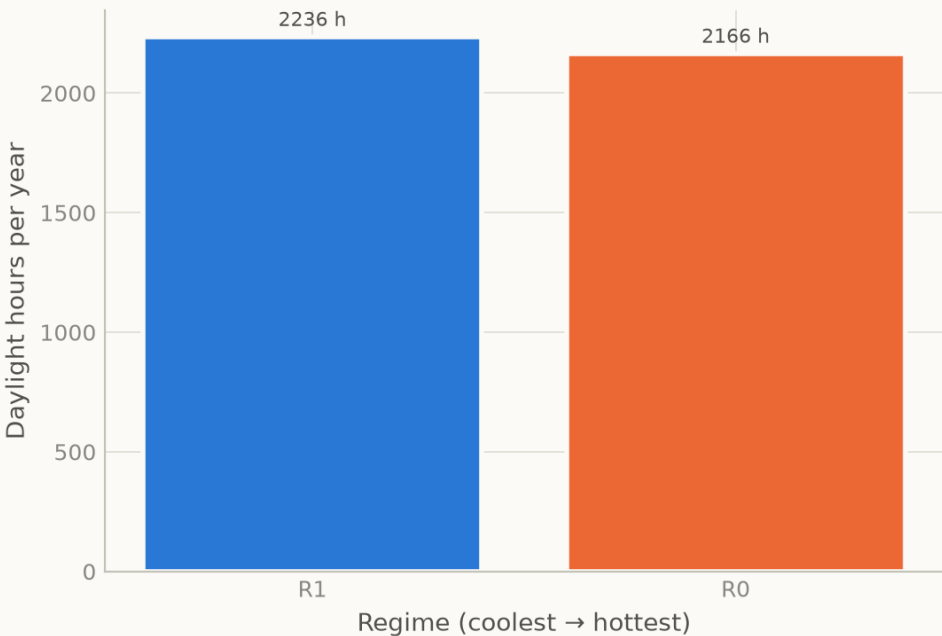
Every path above is a live link. Nothing on this sheet was drawn by hand — regenerate it with `python run_analysis.py`

Fig08 microclimate regimes

The microclimate regimes K-Means separates the site into, and why k = 2

The park's 2 operating regimes, found by clustering

k selected by silhouette score, not chosen — daylight hours only



Each regime becomes a programming slot in the activation strategy.
Source: This project's models — see notebooks/ and src/ · AI Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

DRAWN BY [src/models.py](#) [src/dataset.py](#) READ FROM [data/processed/spatial_grid_comfort.csv](#)

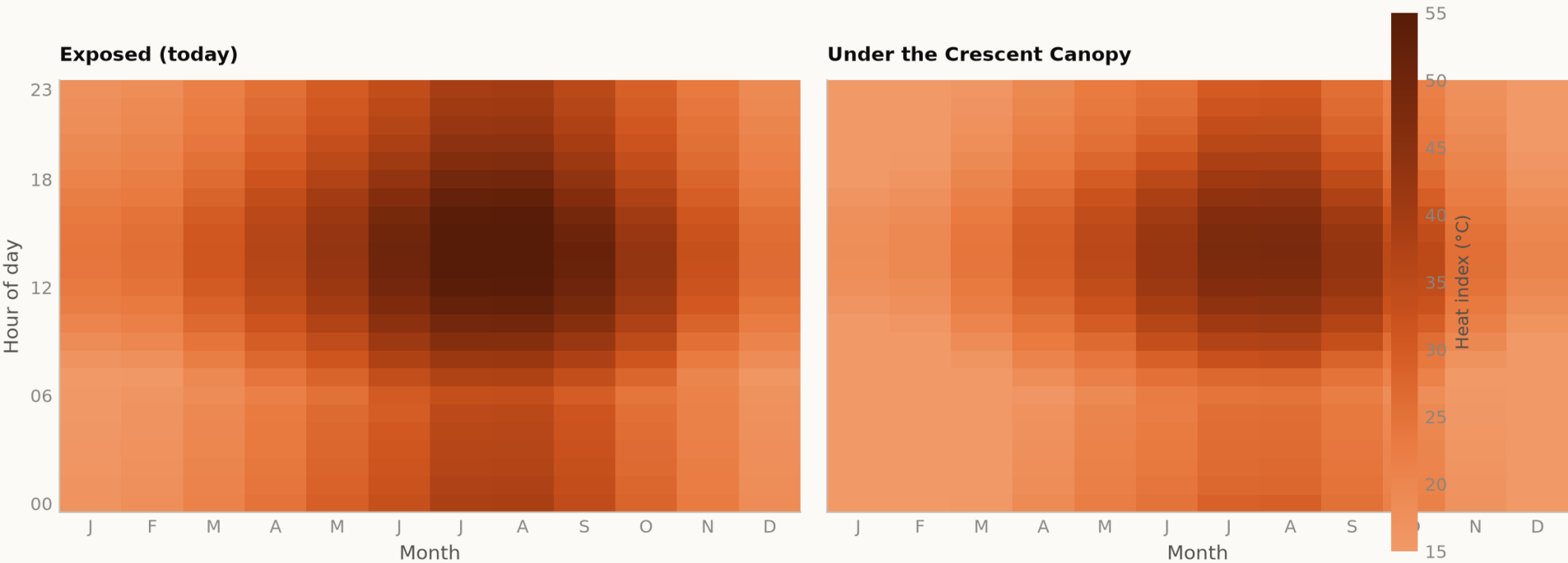
Every path above is a live link. Nothing on this sheet was drawn by hand — regenerate it with python run_analysis.py

Fig09 diurnal comfort

Comfort hour by hour and month by month — where the programme targets are taken from

When is the park comfortable?

Mean heat index by hour and month — exposed today, shaded as designed



The design opens up the late afternoon in spring and autumn, which is when the catchment most wants to use the park.
Source: NCM Dubai climate normals 1977-2015 (39-year record) — hourly series MODELLED, see [src/climate.py](#) · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

DRAWN BY [src/climate.py](#) [src/figures.py](#) **READ FROM** [data/processed/hourly_climate_comfort_8760.csv](#)

Every path above is a live link. Nothing on this sheet was drawn by hand — regenerate it with python `run_analysis.py`

Every step of the work, with its proof attached

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out.
Every file name is a live link — click it and read the actual thing, do not take this document's word for it.

PHASE 8 User Experience & Activation

THIS DOCUMENT RESTS ON



fig08_microclimate_regimes.png

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

CODE [src/models.py](#) [src/dataset.py](#)

PROOF — click to open [data/processed/spatial_grid_comfort.csv](#)

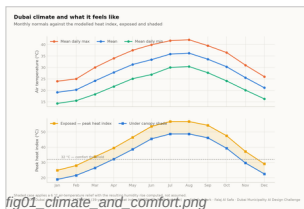


fig01_climate_and_comfort.png

PHASE 1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

CODE [src/climate.py](#) [src/solar.py](#)

PROOF — click to open [data/processed/hourly_climate_comfort_8760.csv](#) [data/raw/sources.json](#)

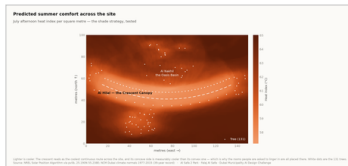


fig04_site_comfort_map.png

PHASE 2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

CODE [src/climate.py](#)

PROOF — click to open [models/headline_metrics.json](#)

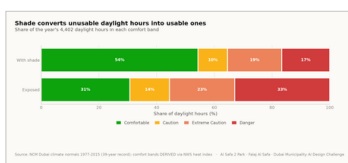


fig02_comfort_bands.png

PHASE 3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

CODE [src/config.py](#) [src/plan.py](#)

PROOF — click to open [models/headline_metrics.json](#)

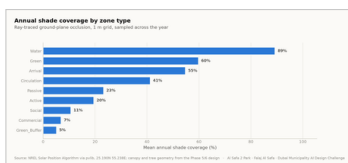


fig03_shade_by_zone.png

PHASE 4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

CODE [src/plan.py](#) [src/solar.py](#)

PROOF — click to open [data/processed/masterplan_geometry.json](#)

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.

Masterplan – Early Site Info

On the site, and using the data of the vicinity, Brian Shaw explains just the same generating the master plan.

fig10_masterplan.png

PROOF — click to open [data/raw/site_zoning_schedule.csv](#)
[data/processed/masterplan_geometry.json](#)

Section A-A – the Crescent Canopy at midspan

Every dimension used here follows ASCE/SEI 2007 and was approved by the WES, September 13, 2007.

The diagram shows a cross-section of the bridge deck and the crescent-shaped canopy above it. The bridge deck is represented by a horizontal line with a depth of 10 ft. The canopy starts 15 ft before the left abutment and ends 15 ft after the right abutment, resulting in a total width of 60 ft between the abutments. The canopy's height varies along its length, reaching a maximum of 18 ft at the center (midspan) and decreasing towards the edges. The bottom edge of the canopy is at a constant height of 10 ft from the deck level. The distance from the centerline of the bridge deck to the centerline of the canopy at midspan is 30 ft.

Legend:

- Top of canopy - 18 ft
- Bottom of canopy - 10 ft

Dimensions:

- Total width = 60 ft
- Span = 30 ft
- Deck depth = 10 ft
- Canopy height at midspan = 18 ft
- Canopy height at edges = 10 ft

Note: The canopy extends 15 ft beyond each abutment.

section_crescent.png

PROOF — click to open [data/processed/planting_layout.csv](#)

The scheme costs £65.37M at M of the £65.37M budget

Capital cost plan - 17% of the budget, £10.00M of investment

Category	Value (£M)
At risk -	10.00
International	10.00
Services and support	10.00
Personnel	10.00
Equipment	10.00
Transport and travel	10.00
Security services	10.00
Training costs	10.00

£65.37M

fig11_cost_plan.png

PROOF — click to open [data/processed/cost_plan.csv](#) [models/cost_summary.json](#)

Does the surrogate actually reproduce the simulation?
 Second Simulacra predictions against the observed ground truth. Note that both axes are in hours.

Left Plot: Observed (hours) vs. Ray-traced whale foraging (hours). $R^2 = 0.9944$. The data points are tightly clustered along the diagonal line, indicating a very high correlation between observed and predicted values.

Right Plot: Observed (hours) vs. Predicted (hours). $RMSE = 87.6$ hours. The data points are widely scattered, indicating a poor fit and high prediction error.

Reproduced with permission from the authors of the paper: "A Surrogate Model for Predicting the Foraging Behavior of Whales" by S. J. Gray, J. H. Wang, & S. J. Gray. Under Microscopy in Design Challenge.

fig05_surrogate_performance.png

PROOF — click to open [models/model_metrics.json](#) [tests/test_pipeline.py](#)

The screenshot displays the JALSA 3.0 Home interface. At the top, the title bar reads "JALSA 3.0 Home - JALSA 3.0. JALSA". Below the title bar, there is a navigation menu with options: "Home", "About JALSA", "Getting Started", "Data", "Analysis", "Visualization", "Help", and "Contact". The main content area is divided into several sections:

- Top Section:** A large orange circular visualization with a central bright spot and surrounding concentric rings, labeled "JALSA 3.0 Home" and "JALSA 3.0. JALSA".
- Left Sidebar:** A vertical list of links: "Home", "About JALSA", "Getting Started", "Data", "Analysis", "Visualization", "Help", and "Contact".
- Right Section:** A series of smaller visualizations and charts:
 - A horizontal bar chart titled "JALSA 3.0 Home" with a legend showing "JALSA 3.0. JALSA" and "JALSA 3.0. JALSA".
 - A line graph titled "JALSA 3.0 Home" showing a trend over time.
 - A heatmap titled "JALSA 3.0 Home" showing a spatial distribution.
 - A small table titled "JALSA 3.0 Home" with columns "JALSA 3.0. JALSA" and "JALSA 3.0. JALSA".

The bottom of the page features a large orange banner with the text "JALSA 3.0 Home" and "JALSA 3.0. JALSA".

board 2_evidence.png

PROOF — click to open [tests/test_pipeline.py](#)

The 4 images in this file, and the whole record

This slot's own pictures come first, larger. The remaining 20 of the 24 images the pipeline produces follow, so the whole visual record is in every file. Each links to its full-resolution original; nothing here is hand-drawn.

01

IN THIS FILE

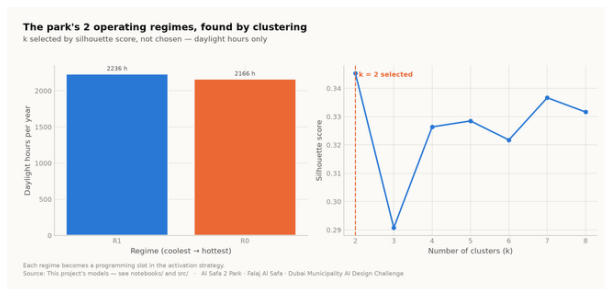


Fig08 microclimate regimes

analysis output

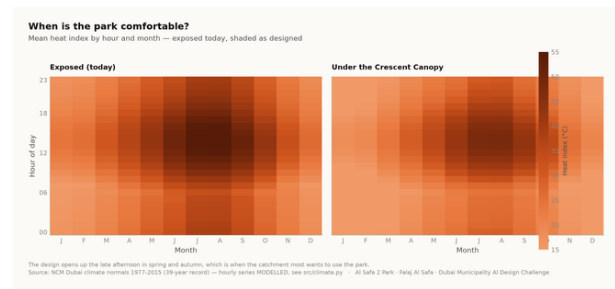
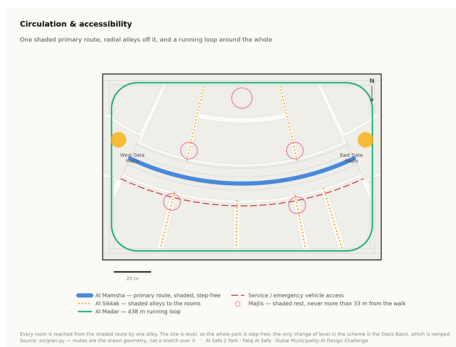


Fig09 diurnal comfort

analysis output



Circulation

technical drawing



Childrens dune play

artistic impression

02

EVERY OTHER IMAGE IN THE PROJECT

The other 20 images the project produces are not reprinted here — this file shows its own. Each name below opens the full-resolution original in the repository.

Fig01 climate and comfort analysis output

Fig04 site comfort map analysis output

Fig07 confusion matrix analysis output

Elevation technical drawing

Section technical drawing

Masterplan aerial golden hour artistic impression

Night plaza render artistic impression

Fig02 comfort bands analysis output

Fig05 surrogate performance analysis output

Fig10 masterplan analysis output

Facilities technical drawing

Board 1 concept presentation board

Oasis basin artistic impression

Souk plaza artistic impression

Fig03 shade by zone analysis output

Fig06 feature importance analysis output

Fig11 cost plan analysis output

Planting technical drawing

Board 2 evidence presentation board

Spine corridor interior artistic impression

Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

07

OF 12

THE FIGURES THIS FILE LEADS ON

each one appears in the full tables below, from the same file

64.6%

COMFORTABLE AS DESIGNED

+20.1 pts

COMFORT HOURS GAINED

4,402

DAYLIGHT HOURS MODELLED

97.5%

COMFORT BAND ACCURACY

MEASURED RESULT

models/headline_metrics.json

Annual daylight hours modelled	4,402
Comfortable daylight hours — today	44.5%
Comfortable daylight hours — as designed	64.6%
Mean heat-index reduction under canopy	7.13 °C
Peak heat index, exposed	56.8 °C
Peak heat index, shaded	48.7 °C
Crescent Walk shaded, canopy and louvre	87.3%
Site-wide mean shade	34.1%
Trees planted	131

TRAINED MODELS, ON HELD-OUT TEST SETS

models/model_metrics.json

M1a Random Forest — shade surrogate, test R ²	0.9982
M1b Neural network — deployed surrogate, test R ²	0.9944
M2 Gradient Boosting — comfort band, test accuracy	97.5%
M3 K-Means — regimes selected by silhouette	k = 2
Training / validation / test split	10,500 / 2,250 / 2,250
Random seed, fixed for reproducibility	42

WHAT THE CHECKS PROTECT

each one asserts what would otherwise drift silently

Climate fidelity

the modelled year reproduces the published normals it was built from

No overlaps

no room in the schedule overlaps another

Areas close

every area sums back to 15,000 m²

Budget held

the cost plan stays inside AED 35 M

No leakage

no model can see its own answer in its inputs

RUN IT YOURSELF

```
git clone https://github.com/wasim437/Al_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js    # 64 portal checks
node tests/test_film.js          # every frame of the film
```

The complete project, and where to check it

Every one of the 124 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

THE SOURCES BEHIND THIS FILE

open these first if you want to check this document

src/models.py	Technical drawing — to scale.
src/climate.py	Technical drawing — to scale.
data/processed/hourly_climate_comfort_8760.csv	Technical drawing — to scale.

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Ma...
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Di...
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (6)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
README.md	The design argument, written for a juror
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

THE TECHNICAL DRAWINGS AND BOARDS (7)

design/visuals/circulation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/elevation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/facilities_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/planting_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/section_crescent.png	Technical drawing — generated from src/plan.py
design/boards/board_1_concept.png	Technical drawing — generated from src/plan.py
design/boards/board_2_evidence.png	Technical drawing — generated from src/plan.py

THE ANALYSIS CODE (13)

src/__init__.py	Analysis module
src/boards.py	The two presentation boards
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/config.py	Every constant the project reads
src/costing.py	The cost model against the AED 35 M ceiling
src/dataset.py	Assembles the ML training tables
src/drawings.py	Section, elevation, circulation, planting, facilities
src/figures.py	The analysis charts
src/models.py	The four models
src/plan.py	SINGLE SOURCE of the crescent geometry
src/solar.py	Sun position and shadow ray-tracing
src/viz.py	Analysis module
run_analysis.py	Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (16)

tools/__init__.py
tools/build_docs.py
tools/build_links.py
tools/build_notebook.py
tools/build_render_film.py
tools/build_reports.py
tools/build_submission_pdfs.py
tools/check_renders.py
tools/embed_narration.py
tools/make_video_mp4.py
tools/make_winning_video.py
tools/organize_repo.py
tools/report_content.py
tools/sync_film.py
tools/sync_portal.py
tools/sync_submission.py

THE DATA, AS ISSUED (7)

data/raw/construction_unit_rates_aed.csv	Source dataset
data/raw/dubai_climate_normals_ncm.csv	Source dataset
data/raw/dubai_population_dsc_2023.csv	Source dataset

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

The complete project — continued

data/raw/site_zoning_schedule.csv	The measured room schedule
data/raw/sources.json	Every source dataset, with its period
data/raw/species_water_carbon_rates.csv	Source dataset
data/raw/walk_catchment_rings.csv	Source dataset

THE DATA, AS PROCESSED (6)

data/processed/cost_plan.csv	The capital cost plan, line by line
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/masterplan_geometry.json	The plan geometry, as exported
data/processed/planting_layout.csv	Every one of the 131 trees
data/processed/schema.json	Generated by the pipeline
data/processed/spatial_grid_comfort.csv	The 15,000-cell ground grid

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	41 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFa_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded
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THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Ai Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file

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The complete project — continued

[submission/09_Material_Landscape_Palette/MANIFEST.md](#)
[submission/10_Complete_Design_Report/MANIFEST.md](#)
[submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md](#)
[submission/12_Concept_Animation_Video/MANIFEST.md](#)

What is in the slot, and what produced each file
What is in the slot, and what produced each file
What is in the slot, and what produced each file
What is in the slot, and what produced each file

WORKING HISTORY (1)

[archive/withdrawn_visuals/README.md](#)

Images withdrawn on purpose, and why